

FAIRBANKS 11 NE, AK, USA

WMO: 702630

Lat: 64.9735N

Lon: 147.5100W

Elev: 1140

StdP: 14.10

Time Zone: -9.00 (NAK)

Period: 02-19

WBAN: 26494

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.2	(c) -21.9	(d) -32.1	(e) 0.9	(f) -25.7	(g) -28.1	(h) 1.2	(i) -20.8	(j) 13.6	(k) 10.2	(l) 11.9	(m) 10.5	(n) 0.2	(o) N/A	(p) 0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	77.3	57.8	74.3	57.0	71.1	55.8	60.4	72.9	58.9	70.1	57.4	67.3	4.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.0	69.6	63.2	54.3	65.3	61.1	52.7	61.6	59.9	27.2	73.2	26.1	70.2	25.1	67.6	66.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.8	8.9	7.7	-30.1	82.7	5.8	2.9	-34.3	84.8	-37.7	86.5	-40.9	88.1	-45.1	90.3
			-29.0	63.4	5.7	2.0	-33.1	64.9	-36.4	66.0	-39.6	67.2	-43.8	68.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.1	4.0	12.1	15.8	33.9	49.0	58.1	59.9	54.9	45.1	30.1	12.2	8.4
	DBStd	23.06	17.22	14.86	13.56	10.52	8.34	6.33	5.48	6.35	7.96	10.30	13.17	13.90
	HDD50	7407	1427	1060	1059	485	119	8	3	21	182	619	1135	1289
	HDD65	12059	1892	1480	1524	932	497	223	175	320	596	1081	1585	1754
	CDD50	865	0	0	0	4	89	251	310	173	35	3	0	0
	CDD65	41	0	0	0	0	2	16	16	7	0	0	0	0
	CDH74	283	0	0	0	0	15	113	109	46	0	0	0	0
	CDH80	25	0	0	0	0	0	13	9	3	0	0	0	0
(14) Wind	WSAvg	2.8	2.0	2.4	3.3	3.7	3.7	3.3	3.3	3.0	3.1	2.6	2.0	1.7
(15) Precipitation	PrecAvg	12.7	0.6	0.5	0.4	0.4	0.7	1.7	2.8	2.2	1.5	0.7	0.7	0.7
	PrecMax	18.6	2.5	1.6	2.1	2.8	2.1	3.1	5.8	3.7	3.0	1.5	1.8	3.0
	PrecMin	8.7	0.1	0.1	0.0	0.0	0.2	0.6	0.2	0.4	0.3	0.2	0.1	0.0
	PrecStd	2.4	0.6	0.4	0.4	0.6	0.4	0.7	1.5	0.9	0.7	0.3	0.6	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.8	44.1	47.9	63.2	76.0	81.9	81.3	78.5	66.7	61.6	43.0	42.3
		MCWB	39.0	35.9	37.7	45.6	53.3	59.5	60.0	58.8	51.9	42.5	35.0	34.0
	2%	DB	38.5	39.2	42.4	56.3	71.0	77.3	77.0	74.5	63.2	51.5	38.9	37.5
		MCWB	32.4	31.8	34.0	42.3	51.6	57.0	59.6	57.9	50.1	41.3	32.6	31.3
	5%	DB	34.2	36.2	38.7	52.2	67.1	73.6	74.1	69.3	60.1	47.6	34.8	33.2
		MCWB	29.7	30.1	31.3	40.1	50.1	56.1	58.7	56.0	48.5	39.1	31.1	28.6
	10%	DB	28.2	32.6	34.6	48.5	63.3	70.3	70.9	65.6	57.1	44.0	30.5	29.0
		MCWB	25.1	27.4	29.1	38.1	48.1	54.6	57.3	54.3	47.0	36.8	27.8	26.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	34.5	39.0	45.1	55.3	62.4	63.2	60.8	53.2	43.9	34.9	34.4
		MCDB	45.4	38.5	48.8	56.5	71.6	75.9	74.9	72.0	62.9	53.0	38.7	40.2
	2%	WB	33.8	31.4	35.8	41.3	53.2	60.0	61.0	58.3	51.2	41.9	32.1	32.2
		MCDB	38.3	37.7	42.9	51.7	69.5	73.1	73.6	67.2	59.9	50.0	36.4	37.7
	5%	WB	30.5	28.8	33.4	39.4	51.3	58.0	59.5	56.6	49.9	39.4	29.7	29.4
		MCDB	35.3	33.8	39.8	49.7	65.8	70.3	71.2	64.7	58.0	46.6	32.9	33.7
	10%	WB	27.6	26.3	30.6	38.0	49.4	56.1	58.2	55.1	48.4	37.4	26.6	26.5
		MCDB	32.0	30.5	36.4	47.4	62.3	67.3	68.5	62.6	56.0	43.1	28.8	29.9
(35) Mean Daily Temperature Range	MDBR		10.8	12.2	14.3	15.2	17.1	16.5	15.0	14.2	12.9	9.9	9.8	11.0
	5% DB	MCDBR	13.1	12.7	15.9	18.6	21.2	20.3	19.8	19.1	17.1	13.4	12.0	14.4
		MCWBR	9.0	9.0	10.0	9.9	9.3	8.9	8.5	8.1	8.5	7.7	8.8	10.0
	5% WB	MCDBR	13.2	12.5	15.2	17.1	20.0	17.7	17.2	15.4	14.9	12.4	11.9	13.0
		MCWBR	9.2	9.1	9.8	9.3	8.9	8.3	7.9	7.7	8.1	7.3	8.9	9.6
(40) Clear-Sky Solar Irradiance	taub		0.164	0.238	0.246	0.297	0.333	0.367	0.367	0.380	0.303	0.254	N/A	0.170
	taud		1.938	2.218	2.308	2.296	2.445	2.360	2.384	2.344	2.546	2.455	N/A	1.940
	Ebn at Noon		124	217	272	279	278	268	265	247	252	215	N/A	36
	Edh at Noon		7	16	24	31	30	34	32	30	20	13	N/A	2
(44) All-Sky Solar Radiation	RadAvg		49	267	840	1456	1661	1745	1509	1147	723	298	74	13
	RadStd		4	21	52	91	125	147	109	115	87	34	9	1

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	+1.55	N/A	N/A	N/A	N/A	N/A	-587	-637	N/A	N/A

Nomenclature: See separate page